

RESEARCH ARTICLE

Alzheimer's Disease Sequencing Project release 4 whole genome sequencing dataset

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Abstract

INTRODUCTION: The Alzheimer's Disease Sequencing Project (ADSP) is a national initiative to understand the genetic architecture of Alzheimer's disease and related dementias (ADRD) by integrating whole genome sequencing (WGS) with other genetic, phenotypic, and harmonized datasets from diverse populations.

METHODS: The Genome Center for Alzheimer's Disease (GCAD) uniformly processed WGS from 36,361 ADSP samples, including 35,014 genetically unique participants of which 45% are from non-European ancestry, across 17 cohorts in 14 countries in this fourth release (R4).

RESULTS: This sequencing effort identified 387 million bi-allelic variants, 42 million short insertions/deletions, and 6.8 million structural variants. Annotations and quality control data are available for all variants and samples. Additionally, detailed phenotypes from 15,927 participants across 10 domains are also provided. A linkage disequilibrium panel was created using unrelated AD cases and controls.

DISCUSSION: Researchers can access and analyze the genetic data via the National Institute on Aging Genetics of Alzheimer's Disease Data Storage Site (NIAGADS) Data Sharing Service, the VariXam, or NIAGADS GenomicsDB.

KEYWORDS

Alzheimer's disease, diversity, genetic architecture, genetics data sharing, genetics knowledge-base, linkage disequilibrium reference panel, whole genome sequencing

Highlights

- We detailed the genetic architecture and quality of the Alzheimer's Disease Sequencing Project release 4 whole genome sequences.
- We identified 435 million single nucleotide polymorphisms, insertions and deletions, and structural variants from diverse genomes.
- We harmonized extensive phenotypes, linkage disequilibrium reference panel on subset of samples.
- Data is publicly available at NIAGADS Data Storage Site, variants and annotations are browsable on two different websites.

1 | BACKGROUND

Alzheimer's disease (AD) is a neurodegenerative condition characterized by the abnormal buildup of amyloid beta peptides in extracellular plaques and hyperphosphorylated tau in intracellular neurofibrillary tangles. This progressive neurodegeneration results in a gradual decline in cognitive and functional abilities. Genetic variants play a significant role in the development of late-onset AD (LOAD). The first notable finding in 1993 reported the $\epsilon 4$ allele of the apolipoprotein E (APOE) gene was associated with the risk of developing AD.¹ The identification of additional genetic factors for LOAD accelerated with the advent of high-throughput genomic technologies, such as genotype arrays, in the late 2000s.^{2,3} Since then, the list of genetic factors continues to expand with the inclusion of larger sample sizes through international consortium efforts, notably between the Alzheimer's Dis-

ease Genetics Consortium (ADGC) and the International Genomics of Alzheimer's Project (IGAP).⁴⁻⁶ However, most of these identified genetic variants are common alleles with individually small causal effects on disease susceptibility as the studies focused on genotype array data. These variants contribute minimally to the overall genetic liability for the disease, as a study showed that the single nucleotide polymorphism (SNP) heritability estimate from the largest AD genome-wide association studies (GWAS) to date is 3.1%,⁷ which is significantly smaller than the heritability estimates for AD obtained from twin studies, which ranges from 60% to 80%.^{8,9}

Whole-genome sequencing (WGS) can address this missing heritability challenge by (1) providing a more comprehensive view of the genetic architecture via a broad range of variants, and (2) identifying rare variants with potentially larger phenotypic effects. Analyzing many samples is necessary to address the above gaps. WGS studies are

more costly compared to genotyping array, and as a result fewer participants of non-European descent have been sequenced.¹⁰ Expanding AD research to ancestrally diverse populations is crucial for several reasons.

Most genetic studies of AD have focused on non-Hispanic White populations. However, genetic risk factors identified in non-Hispanic White populations may not fully explain the observed ethnic disparities in AD. For instance, while *APOE ε4* is a significant predictor of late-onset AD in non-Hispanic White individuals, its predictive power is weaker and more inconsistent in Black Americans, and Hispanic or Latino populations.^{11–13} Investigating how genetic risk factors for AD vary among ethnic groups could pave the way for more effective, tailored treatments and interventions. Notably, certain genetic variants—such as those in *SORL1*, *ABCA7*, and *ACE*—exhibit stronger associations with AD risk in specific groups, including Asians,¹⁴ Black Americans,¹⁵ and Israeli-Arabs.¹⁶ These findings suggest that therapies targeting these genes may have varying levels of effectiveness across different ethnicities.

Funded through cooperative agreements and research grants, the Alzheimer's Disease Sequencing Project (ADSP) brings together 497 investigators from institutions worldwide. In 2023, it successfully completed the "Follow-Up Study (FUS) Phase" (the third phase of ADSP), sequencing existing cohorts of Black Americans and those of pan-Hispanic ancestry at the American Genome Center at the Uniformed Services University of the Health Sciences (USUHS) and John P. Hussman Institute for Human Genomics (HIHG). This effort was conducted in collaboration with established National Institutes of Health (NIH)-funded AD infrastructure like the National Centralized Repository for Alzheimer's Disease (NCRAD), National Institute on Aging Genetics of Alzheimer's Disease Data Storage Site (NIAGADS), and the Genome Center for Alzheimer's Disease (GCAD). Additionally, participants from non-Hispanic White and Asian ancestries have been sequenced. All genomes, including those from previous ADSP phases and other collaborative projects, have been processed using a unified pipeline, subjected to comprehensive quality control, and annotated using various resources. This collection, the release 4 (R4) of ADSP data, forms the world's largest publicly available AD genome resource.

2 | METHODS

2.1 | Sequence existing ancestrally diverse cohorts via the ADSP FUS

The ADSP FUS is a National Institute on Aging (NIA) initiative focused on identifying genetic risk and protective variants for AD by expanding the ADSP cohorts beyond primarily participants with non-Hispanic Whites of European Ancestry (<https://adsp.niagads.org/adsp-and-affiliates-whole-genome-sequencing-report/>). Given the lack of ancestral diversity in the ADSP, the ADSP FUS was designed to sequence existing ancestrally diverse and unique cohorts. The current phase for ADSP FUS, ADSP FUS 2.0: The Diverse Population Initiative, focuses on Hispanic or Latino populations, non-Hispanic Black with

RESEARCH IN CONTEXT

- 1. Systematic review:** The Alzheimer's Disease Sequencing Project release 4 dataset represents the most extensive and diverse genetic resource for Alzheimer's disease (AD) research, comprising 36,361 whole-genome sequences, including 35,014 unique participants with significant representation from non-European ancestries. It identifies > 387 million genetic variants and integrates phenotypic data to explore the genetic architecture of AD. Key contributions include filling gaps in studies of underrepresented populations, advancing ancestry-specific analyses, and annotating rare and common variants.
- 2. Interpretation:** The dataset can uncover new population-specific genetic risk factors. The uniformly processed and harmonized sequencing and phenotypic data can ensure high utility for cross-cohort analyses. However, challenges like population imbalances need addressing for more reliable interpretations.
- 3. Future directions:** Future work should focus on increasing sequencing in underrepresented populations, refining analytic pipelines for rare variants, and developing genetic insights-based targeted therapies. Expanding access through platforms like the National Institute on Aging Genetics of Alzheimer's Disease Data Storage Site and VariXam will further enhance the dataset's global impact on AD research.

African ancestry, and Asian populations (e.g., the Asian cohort for AD). The ADSP FUS initiatives intend to sequence > 100,000 participants from diverse ancestries. See [Methods S1](#) in supporting information for cohort descriptions. Workflows within the FUS infrastructure (Figure [S1](#) in supporting information) include biospecimens being processed and DNA prepared and allocated for WGS at USUHS and HIHG. All raw sequence data are transferred to the GCAD for processing and harmonization after quality control (QC) analysis at the University of Pennsylvania and University of Miami, resulting in analysis-ready genotype and sequence data. All clinical, genotype, and sequence data are housed at the NIAGADS, which stores, manages, and distributes ADSP FUS data to AD researchers.

2.2 | Dataflow and sample processing protocol on single nucleotide variants and insertions and deletions

The data flow from receiving data to data sharing is shown in Figure [S2](#) in supporting information. NIAGADS receives genomes in FASTQ, Binary Alignment Map (BAM), or CRAM (compressed version of BAM) formats from national or international investigators through secure file

transfer protocol or Amazon S3 bucket. NIAGADS also received the identification (ID) information to generate a unique ADSP ID, companion array data for checking concordance, ADSP minimal phenotypes, Institutional Certification forms, and cohort and study information for these genomes. Once the unique ADSP IDs are received and generated by NIAGADS, GCAD can begin production on the genomes.

All genomes are processed using the VCPA pipeline¹⁷ (<https://bitbucket.org/NIAGADS/vcpa-pipeline/src/master/>), a pipeline optimized for the large-scale production of WGS in an Amazon cloud environment and including all steps from aligning raw sequence reads to variant calling using Genome Analysis Toolkit (GATK) best practices. Depending on the file type received, the first few steps of the pipeline are slightly different:

1. CRAM as input: decompression to BAMs is required before running the pipeline as if the input is in BAM format
2. BAM as input: roll back to uBAM (unmapped BAM, i.e., FASTQ) format before running the pipeline as if the input is in FASTQ format
3. FASTQ as input: no special steps

Using VCPA, FASTQs were first mapped/remapped to hg38 reference genome using BWA-mem (v0.7.15). Duplicated reads were then marked using BamUtil (v1.0.13). Base recalibration and insertions and deletions (indel) realignment were done by GATK (v4.1.1) and a CRAM that contains all the above information was stored. Genotype calling was then done by GATK4.1.1, resulting in one genomic variant call format (gVCF) across all chromosomes (1-22, X, Y, and M).

To join genotype single nucleotide variants (SNVs) and short indels, we followed the steps in GATK best practices (version GATK4.1.1) to create a joint genotyped called variant call format (VCFs) using gVCFs generated in step 2. The VQSR model was performed for all chromosomes (1-22, X, Y, and M). This is the "Preview VCF."

2.3 | GRAF-pop

GRAF-pop offers a novel method for global ancestry inference,¹⁸ distinguishing itself from traditional approaches that require genotypes from reference populations or study participants to determine an individual's ancestry. Instead, GRAF-pop precomputes allele frequencies for selected SNPs across reference populations and infers ancestry by directly comparing the test genotypes to these predefined frequencies, eliminating the need for other individual genotypes.

2.4 | Population substructure

We included bi-allelic variants that have (1) passed ADSP QC, that is, with variant flag = 0, (2) allelic read ratio > 0.3 or < 0.7, (3) minor allele frequency (MAF) > 0.02, (4) call rate > 99.5% and with Hardy-Weinberg test *P* value > 0.0005. Then we applied linkage disequilibrium (LD) pruning with R^2 cutoff at 0.05 and window size of 500K base pairs (bp). Across all chromosomes, 146,964 pruned variants

remained for principal components (PCs)/genetic relationship matrix (GRM) calculations. We used R package "GENESIS" (2.20.1)¹⁹ along with GWASTools (1.36.0)²⁰ and SNPRelate (1.24.0)^{21,22} to calculate the PCs and GRM. Five thousand five hundred thirteen out of 36,361 subjects were grouped as related at default threshold. Note, the threshold value on kinobj used for declaring each pair of participants as related or unrelated: the default value is $2^{-(11/2)} \sim 0.022$, corresponding to fourth-degree relatives.

gnomAD 4.0 genome site data provided allele frequency at ancestry level, which can be used for ancestry inference. There are 10 ancestries in the data: African/African American, Amish, Latino, Ashkenazi Jewish, East Asian, Finnish, Middle Eastern, non-Finnish European, Remaining, and South Asian. One hundred forty-five thousand two hundred seventy-eight variants were matched between the ADSP pruned file and gnomAD4.0 and passed to Software SCOPE (<https://github.com/sriramlab/SCOPE.git>) to perform the ancestry proportion analysis. The subject was assigned to the ancestry group in which it has the highest ancestry proportion value.

2.5 | ADSP sample level QC protocol

Four checks were performed to identify potentially low-quality samples for exclusion from the VCFs: (1) SNV concordance check with available GWAS genotypes; (2) sex check for possible sample swaps or misreporting; (3) contamination check for possible sample swaps; and (4) relatedness check to confirm known relationships, identify unknown genetically identical duplicates, and assess potential cryptic relatedness.

The SNV concordance check was done by comparing genotypes taken from existing GWAS array data and genotypes from the pre-view project level VCFs (pVCFs) on all overlapping genotypes. The goal of this analysis is to ensure that samples and IDs match throughout the data management and calling processes. First, GWAS datasets were converted into VCF format. Then GATK was used to compare the pVCF and GWAS file genotypes using the following parameters: `Java -jar GenomeAnalysisTK.jar -T GenotypeConcordance -R hg38.fa -eval [PVCF] -comp [GWAS_VCF] -o [OUTFILE]` where [GWAS_VCF] is the VCF file converted from GWAS, [PVCF] is the Preview pVCF, and [OUTFILE] is the user-selected output filename. Of all samples kept, they all have a concordance of > 0.91.

Sex checks were performed using BCFtools,²³ VCFtools,²⁴ and PLINK²⁵ with the following steps:

1. Use BCFtools to convert chromosome X pVCF into PLINK format
2. Filter out chromosome X pseudo-autosomal region (PAR)
3. Filter out SNVs with MAF < 0.05, run "impute-sex"
4. Run "sex-check" using PLINK for comparison

Comparing the results of "impute-sex" in BCFtools and "-sex-check" in PLINK without filtering on MAFs or without excluding variants in the PAR, the findings of the two approaches are the same. There are 1918 samples with *F* statistic values between 0.2 and 0.8, but only

392 between 0.3 and 0.7. A total of 74 samples were identified to be the incorrect sex. The submitting centers agreed that these subjects needed to be dropped.

Sample-specific contamination checking was performed by using VerifyBamID²⁶ to calculate the concordance estimate between the GWAS genotype data and the BAM file. This approach provides information that can be interpreted to identify potential sample contamination or swapping using the GWAS-BAM contamination estimate. The “FREEMIX” modeling approach was used in this analysis. We used suggested thresholds for contamination taken from the VerifyBamID website: a sample is potentially contaminated if the FREEMIX value is > 0.05 . The command line to generate FREEMIX contamination estimates value is as follows: `verifyBamID -vcf [INPUT.VCF] -bam [INPUT.BAM] -out [OUTPUT.PREFIX] -verbose -ignoreRG` where [INPUT.VCF] is the VCF file converted from GWAS, [INPUT.BAM] is the BAM file generated using VCPA1.1, and [OUTPUT.PREFIX] is the user-selected prefix to be appended to output files. As implemented, DNA sample contamination is indicated if the FREEMIX value is > 0.05 . Across all samples, all FREEMIX values obtained are < 0.05 .

Relatedness checks were performed using PLINK as described:

1. Convert *.vcf.gz files into PLINK binary format—Run PLINK “-recode” and “-make-bed” commands to convert “*.vcf.gz” files into “*.bed”, “*.bim”, and “*.fam” files
2. Combine 22 chromosome-specific PLINK filesets into one genome-wide set
3. Clean and rename empty name “.” snvs to “chr_pos” format to combine
4. Run PLINK “-merge-list” and “-make-bed” commands to combine 22 binary files into one
5. Run PLINK “-extract” command to extract only overlapping ~21k SNVs
6. Run PLINK to generate pairwise sample identity-by-descent (IBD) values
7. Run PLINK “-genome” command on combined file with ~21k SNVs to get pairwise IBD values
8. Using these IBD values, identify related pairs with $IBD_{PI_HAT} > 0.4$

The PI_HAT measurement (Proportion IBD, i.e., $P[IBD = 2] + 0.5 \cdot P[IBD = 1]$) was reported for each pair of samples. All pairs with $PI_HAT > 0.4$ were evaluated for known relatedness.

There was a total of 1205 genetically unique participants identified with multiple samples (including five pairs of twins) across the R4 dataset. This included six known replicates (sequenced a total of 67 times as technical replicates), and 1199 unintentional replicates (1173 pairs and 26 triplicates). After removing the ADSP recommended duplicate samples, there were a total of 35,023 genetically unique samples. (Note that there are 24 subjects from the R4 listed in both the family-based and case/control phenotype files. These samples were either sequenced [1] as part of a case/control study but were also part of a family so their phenotypes are provided in both files, or [2] in both a case/control and family-based study. ADSP recommends using samples from family-based sets.)

After performing sex check, GWAS concordance, and contamination checks for each sample, together with the metrics we collected per sample in VCPA, we designed a list of criteria to drop or fail samples based on this sample level QC process (Figure S3 in supporting information). A sample is dropped if (1) its average genome coverage is < 20 ; or (2) multiple of its non-related sequencing metrics are of bad quality; or (3) it fails contamination check, concordance check, or sex check; or (4) it is an unexpected duplicate. Alternatively, a sample is flagged if fails in less than three sequencing metrics.

2.6 | ADSP variant QC protocol

Different filtering and QC strategies were applied at each level (genotype-, and variant-).²⁷ The QC protocol was applied on the bi-allelic autosomal VCFs. All QC flags were applied uniformly across all samples, regardless of cohort or sequencing information. Variants of low quality have been flagged but these variants have not been excluded and filtered out of the datasets.

1. Genotype-level QC was applied to individual genotypes. Each genotype was evaluated and set to missing (“.”) if either or both read depth (“DP”) was less than 10 ($DP < 10$) or genotype quality (“GQ”) score was less than 20 ($GQ < 20$). All these censored genotypes were excluded from subsequent QC steps, except for estimation of variant-level averaged depth (“AverageReadDepth”) in variant-level QC.
2. Variant-level QC was applied to all variants. Flags were applied in the following order: (a) variants in GATK low sequence quality tranches (variants without a FILTER value of “PASS” that are above the 99.8% VQSR Tranche), (b) monomorphic variants were flagged, (c) variants with high missing rate were flagged, (d) variants with high read depth were flagged.
3. Variants with excessive heterozygosity or departure from Hardy-Weinberg equilibrium (HWE) were evaluated within race/ethnic subgroup; however, given the complexity around race/ethnic subgroups, these were not flagged though the measures have been made available and can be implemented as user-defined filters if desired. Similarly, (“ABhet”) was computed among uncensored heterozygotes at each variant and provided in the files but not applied in any filtering criteria.

2.7 | Comparison to gnomAD

The gnomAD resource (version 4) was download from <https://gnomad.broadinstitute.org/downloads#v4>. This contains bi-allelic variants from multi-ethnic samples. To compare to the ADSP R4 data, we first extracted the bi-allelic variants from the VCF. We then broke the VCFs down into samples of four ancestry groups: non-Hispanic Whites, Hispanics or Latinos, Asians, and non-Hispanic Blacks with African ancestry. Monomorphic variants were excluded from each of the VCFs. We then extracted variants from both the gnomAD and ADSP R4 data

by MAF thresholds: $\leq 0.1\%$, $\leq 0.5\%$, $\leq 1\%$, and $\leq 5\%$, and compared them at the site level. When compared across ancestry groups, only sites that are observed in all ancestries were used for analyses.

2.8 | Variant annotation protocol

All R4 bi-allelic variants have been annotated using the official ADSP annotation pipeline. First, the QCed VCFs were processed using VEP103²⁸ (with the `-everything` flag). Then the JSON-formatted VEP output was processed so that variants affecting multiple transcripts of the same gene were collapsed to generate a “most damaging” consequence for each affected gene. This process uses the ranking table specified in the file “ranking_table.txt” to identify the “most damaging” consequence and to assign an impact score, using a custom annotation ranking process which down-weights consequences for non-sense mediated decay transcripts and non-coding transcripts. Next the QCed VCFs were also processed by SnpEff v5.1d (build 2022-04-19).²⁹ Variants are matched by chromosome, position, reference allele, and alternate allele to CADD v1.6 scores.³⁰ Last, short indels not defined in CADD reference files were processed by CADD and integrated into the dataset. This resource is available at NIAGADS open access <https://dss.niagads.org/open-access-data-portal/>.

2.9 | Loss of function analyses

The functional impact of variants was assessed using SnpEff.²⁹ Variants were annotated as “MODERATE” and “HIGH” when they were protein-altering, while variants with “LOW” and “MODIFIER” effects were considered non-protein altering. All variants categorized as “HIGH” were expected to be disruptive or cause loss of function (LoF) in the protein. Variants with a “MODERATE” effect were expected to be missense and splice region variants. Variants with “MODIFIER” and “LOW” effects were in non-coding regions or were non-disruptive to protein functions. We focused on the LoF variants categorized as “HIGH” in the ADSP cohort.

2.10 | FAVOR annotation protocol

We downloaded the FAVOR³¹ database annotations from <https://docs.genohub.org/data> in July 2023 and used that to annotate all the R4 bi-allelic variants. Then annotation-only VCF files were converted to Genomic Data Structure (GDS) format using the SeqArray package,²¹ containing 156 columns. This resource is available at NIAGADS open access <https://dss.niagads.org/open-access-data-portal/>.

2.11 | Structural variant calling protocol

The GCAD and ADSP structural variant (SV) workgroup together designs the production pipeline which includes Manta³² (v1.6.0) and Smoove (ref, v0.2.6) (<https://github.com/brentp/smoove>) for calling indels. Individual Manta and Smoove callsets were first

merged for each sample and merged with all samples by SVIMMER (v0.1) (<https://github.com/DecodeGenetics/svimmer>). Then, GraphTyper (v2.7)³³ was applied on the merged VCF for SV joint genotyping. Note that the only filter applied is SV size > 10 Mbp; other than that, there is no advance filter. The code used to generate these SV calls is available at: <https://github.com/Illumina/manta>, <https://github.com/brentp/smoove>.

2.12 | LD reference panel

We inferred LD separately for each of the four major ancestral groups (non-Hispanic Whites, Hispanics or Latinos, Asians, and African Americans). All participants for both cases and controls were included but progressive supranuclear palsy (PSP) and corticobasal degeneration (CBD) samples are not included, resulting in 32,236 participants total. We calculated LD for all pairs of variants with minor allele count (MAC) ≥ 5 and within 5 Mbps of each other with emerald³⁴ using the following parameters, `-mac > 5`, `-threshold 0.2`, and `-window 5000000`. Each segment was analyzed by 5Mb window with a 3Mb overlap, then we concatenated all segments and removed duplicate records. Only variant pairs with $R^2 > 0.2$ were then retained. For each variant pair, we reported variant genomic positions, reference and non-reference alleles, their R and R^2 correlation, and D and D' statistics. This resource is available at NIAGADS open access <https://dss.niagads.org/open-access-data-portal/>.

2.13 | ADSP Phenotype Harmonization Consortium

The ADSP Phenotype Harmonization Consortium (ADSP-PHC) harmonizes all available data from each domain, regardless of sequencing status, to ensure the highest quality harmonization. The harmonized phenotypic data are then subset to participants with available WGS. These data are released per participant via NIAGADS. All harmonized data can be accessed directly from each cohort. All ADSP phenotype data are harmonized by a multi-disciplinary team that includes world experts in neuroimaging, neuropsychology, fluid biomarkers, neuropathology, and vascular contributions to AD and related dementias (ADRD). Data processing and domain-specific harmonization protocols are available in [Methods—ADSP Phenotype Harmonization Consortium Protocol](#) in supporting information. These files are available in <https://dss.niagads.org/>.

2.14 | VariXam

VariXam is an aggregated database and a variant browser that shows genomic variants detected on WGS or whole exome sequencing (WES) data of the ADSP. The database currently includes variants of all the R4 36K WGS and allows users to search for genes or variants of interests. The human reference genome used is GRCh38. It is available here: <https://varixam.niagads.org/>.

2.15 | NIAGADS Alzheimer's GenomicsDB

The NIAGADS Alzheimer's GenomicsDB is an interactive knowledge-base for AD genetics.³⁵ The resource provides unrestricted access to GWAS summary statistics datasets, variant annotations, and meta-analysis results deposited at the NIAGADS. The platform allows users to search for genes or variants of interests, and interactively mine or visually inspect datasets and annotated ADSP variant tracks on a genome browser. The GenomicsDB can be accessed at <https://www.niagads.org/genomics>.

2.16 | NIAGADS Data Sharing Service

The NIAGADS Data Sharing Service (DSS) was developed to facilitate the deposition and sharing of whole-genome and whole-exome sequencing data from ADSP and other National Institute on Aging (NIA)-funded ADRD studies to the research community at large. In keeping with the NIH GDS policy, all genomic data are classified as controlled access as outlined in the Institutional Certification forms provided by the submitting institutions. Principal investigators can request DSS-distributed data through the Data Access Request Management (DARM) system by logging in using their eRA Commons ID. Once an application is approved by the NIH-formed NIAGADS ADRD Data Access Committee (NADAC) and Data Use Committee (DUC), the data can be accessed through the Data Portal and downloaded directly or through Amazon EC2. DSS can be found at <https://dss.niagads.org/>.

2.17 | Released genotyping files

Due to the sheer sizes of pVCFs, all pVCFs are split by chromosomes. We provide three versions of pVCFs for users to choose from: (1) "Preview pVCF," (2) "Compact pVCF": only reserved GT (genotype) of each sample for each variant, (3) "Compact filtered pVCF": a compact version with replacing low-quality genotypes to missing (.). Each set of pVCF files are divided by chromosome, then split into bi-allelic and multi-allelic variant files. We also created the fully QCed VCF (output from the "ADSP Variant QC protocol") in GDS format to facilitate analysts using R for downstream association analyses. These files are available in <https://dss.niagads.org/>.

3 | RESULTS

3.1 | Sequence existing ancestrally diverse cohorts via the ADSP FUS

The ADSP FUS is a NIA initiative focused on identifying genetic risk and protective variants for AD by expanding the ADSP cohorts beyond primarily participants with non-Hispanic Whites. Given limitations in population diversity in the ADSP, the ADSP FUS was designed to

sequence existing ancestrally diverse and unique cohorts. ADSP FUS 2.0 focuses on Hispanics and Latino populations, non-Hispanic Blacks with African ancestry, and Asian populations (e.g., the Asian cohort for AD). ADSP -FUS intends to sequence > 100,000 participants from diverse ancestries (Methods S1). ADSP has developed a workflow (Figure S1) to support biospecimens processing, DNA preparation, and sequencing at USUHS and HIHG.

3.2 | Sample characteristics

Twenty thousand seven hundred seventy-one WGS data (8159 new) from the ADSP FUS phases are included in the ADSP R4 dataset, bringing the total number to 36,361 across 17 cohorts/studies in 14 countries (Figure 1A). Sequencing was carried out at 10 sequencing centers using Illumina technology. Most were generated using the polymerase chain reaction (PCR)-free protocol (91%) and 150 bp in read length (94%). Three sequencing platforms were used: Illumina 2000/2500 (7%), HiSeqX (35%), or NovaSeq (58%) machines. Table S1 in supporting information provides more details on sequencing configurations. GCAD processed all 36,361 WGS samples from read mapping to variant calling using a standard pipeline (VCPA1.1) to harmonize all data and minimize batch effects¹⁷ (Methods—Dataflow and Sample processing protocol on SNVs and indels in supporting information).

Of these 36,361 samples, 35,014 participants are genetically unique. Based on ethnicity reported by cohorts, there are 5260 African Americans, 78 American Indian or Alaskan Native, 2787 Asians, 10,972 Hispanics, 9 Native Hawaiian or Pacific Islander, 13,969 non-Hispanic White participants, and 1896 participants of unknown ethnicity. A breakdown of demographic information including age, sex, and APOE alleles, is summarized in Figure 1A and Table S2 in supporting information.

We inferred the genetic ancestry of each individual using called genetic data to investigate the discordance between the reported and genetically inferred ancestries. Most discordances reported were in admixed participants,³⁶ presenting additional challenges in identifying ancestry-specific variants. We used GRAF-pop¹⁸ (Methods—GRAF-POP in supporting information), which assumes that each individual is an admixture of three ancestral groups: European (e), African (f), and Asian (a). GRAF-pop estimates ancestry components P_e , P_f , and P_a , which are then used to assign participants to population groups, including European, African/African American, Latin American, Asian-Pacific Islander, and East/South Asian. Using the software's default settings, the match rates between reported and genetically inferred ancestries were 99.4% for European, 98.3% for African American, 96.8% for Asian, and 80.0% for Hispanic participants (Figure 1B and C).

We also performed PCA based on genotypes derived from WGS (Methods—Population substructure in supporting information). We selected common variants (MAF > 0.02) of high quality and performed LD pruning to yield 146,964 variants, then calculated PCs and GRM. We then performed the ancestry inference analysis using 145,278 vari-

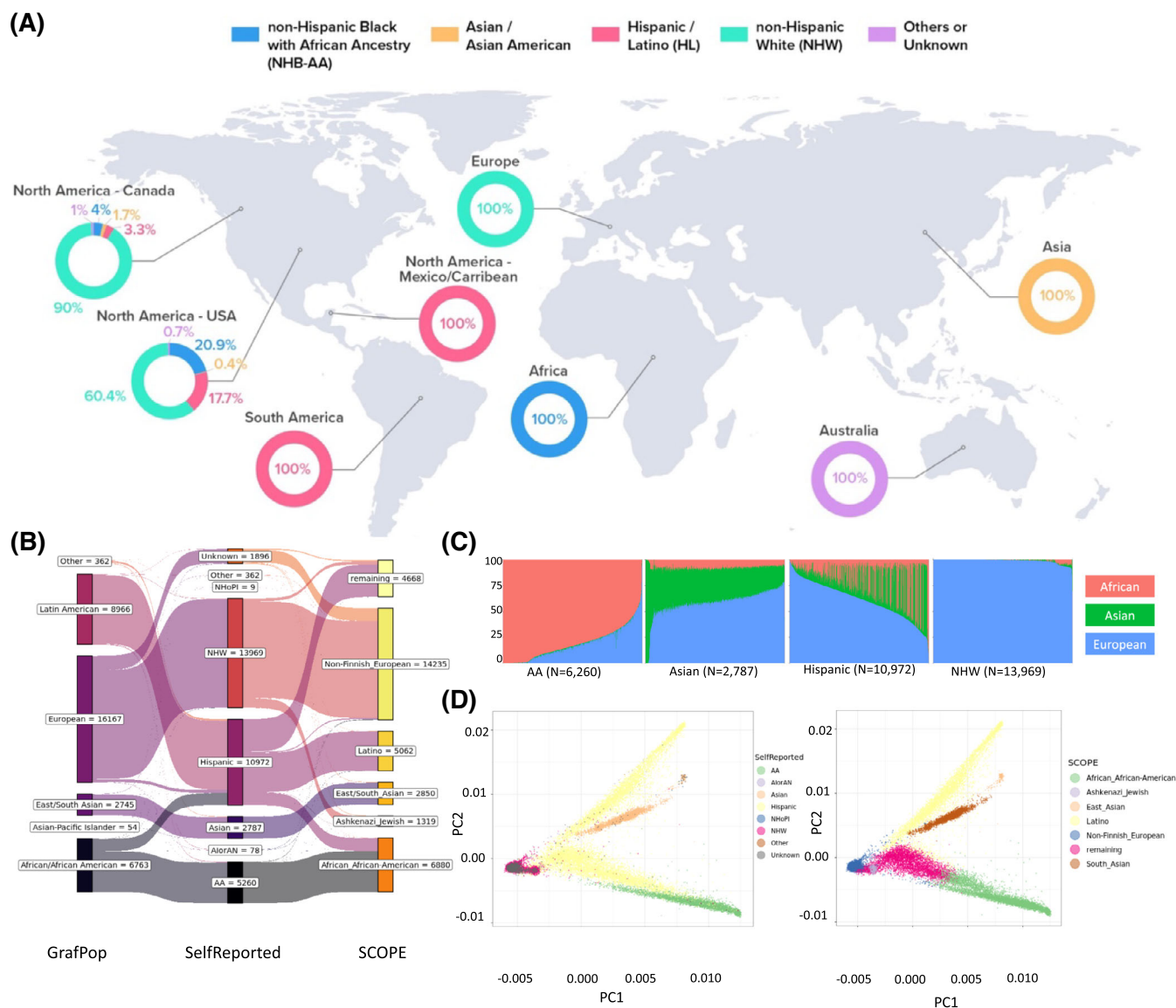


FIGURE 1 Participants in ADSP R4 dataset. A, Worldwide cohorts assembled for this ADSP R4 dataset. Non-Hispanic Black with African Ancestry (NHB-AA) samples are from Africa, and North America (Canada, United States); Asian and Asian American are from Asia, and North America (Canada, United States); Hispanic/Latino (HL) are from North America (Mexico/Caribbean, Canada, United States), and South America; non-Hispanic White (NHW) are from Europe and North America (Canada, United States). Last, some samples categorized as others or unknown and they are from Australia. B, Comparison of reported ethnicity against those inferred by GRAF-pop- and SCOPE-based methods. C, Estimated GRAF-pop ancestral components P_e , P_f , and P_a for all participants. D, PCA plot on R4 participants colored by reported ethnicity (top) or SCOPE. ADSP, Alzheimer's Disease Sequencing Project; PCA, principal component analysis.

ants common in both the ADSP R4 and gnomAD data,³⁷ a publicly available population genetics resource generated on 76,215 diverse samples. Subjects were assigned to an ancestry group in which it has the highest ancestry proportion value. The match rates between reported and genetically inferred ancestries by this method were 88.3% for non-Finnish European and 8.7% for Ashkenazi Jewish; 98.7% for African American; 99.7% for Asian; and among Hispanic participants, 45.5% for Latino, 15% for African American, and 37.8% for remaining. The remaining subjects are most likely from the Caribbean region (Figure 1B and D). Cumulative variance explained by the top 10 PCs is shown in Figure S4a in supporting information.

3.3 | WGS sample quality assessment

We performed QC checks on all samples and 734 samples with quality issues (Method—ADSP sample level QC protocol in supporting information) and reassessed the quality of the callset ($N = 36,361$). The mean read depth across samples is 40.4x with 99% of samples having a coverage $> 30x$ (Figure 2A). The per genome percentage of bases with the quality score greater than Q30 (sequencing error rate $< 0.1\%$) is $90.18 \pm 2.43\%$. On average, $98.92 \pm 2.13\%$ reads of samples are mapped, and $94.14 \pm 2.76\%$ of paired-end reads have both ends mapped.

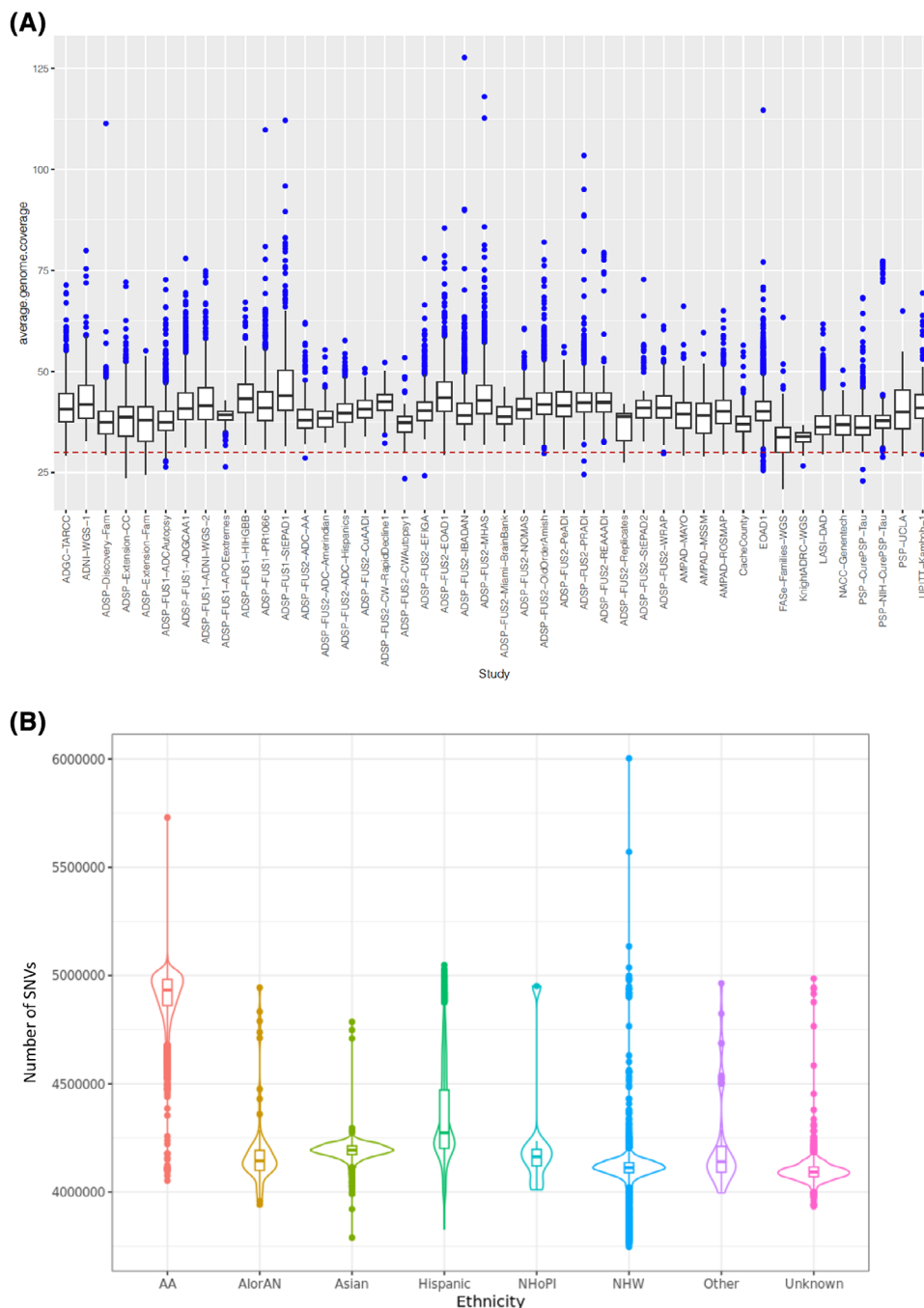


FIGURE 2 WGS sample quality. A, Coverage (30x) for the ADSP R4 data. Red dotted line indicates coverage value at 30.99% of samples pass this threshold. B, Number of SNVs called per sample in each reported ethnic group. Line in each displayed boxplot denotes the mean value where each dot is a sample. ADSP, Alzheimer's Disease Sequencing Project; SNV, single nucleotide variants; WGS, whole genome sequencing.

On average, each sample contained 4.3 million SNVs and 999,000 indels. African American samples have the highest number of variants (4.9 million) followed by Hispanic samples (4.3 million), Asian samples (4.2 million), and non-Hispanic White samples (4.1 million; Figures 2B, Figure S4b).

We assessed if the number of called variants was affected by sequencing configurations such as sequencing platform and use of PCR. We found that samples (regardless of ethnicity) sequenced by the PCRFree protocol tend to yield more variants,

with the combination of NovaSeq + PCRFree returning the highest (Figure S4b, c).

3.4 | Bi-allelic variants (SNVs and indels) in ADSP R4

We called genotypes for all observed variants across all 36,361 samples and split the joint-called results into two VCF file sets. One VCF

TABLE 1 Number of variants (SNVs and indels) identified in the four major ethnic groups in ADSP R4 data, broken down allele frequency (AF). Shown in top is the variant count and percentages per ethnicity. Instead of the total number of variants identified, we showed at the bottom the ethnic specific variants.

All	AA		Asian		Hispanic		NHW	
AF	Count	%	Count	%	Count	%	Count	%
Singleton	48,980,233	48.4	38,123,455	55.0	61,467,493	46.4	78,473,265	57.8
<0.1%	27,252,406	26.9	15,126,076	21.8	48,762,383	36.8	43,759,910	32.3
0.1%–1%	11,674,387	11.5	8,038,369	11.6	12,218,350	9.2	5,653,864	4.2
1%–5%	5,806,358	5.7	2,370,814	3.4	4,054,972	3.1	2,242,076	1.7
>5%	7,514,019	7.4	5,679,746	8.2	5,921,548	4.5	5,543,740	4.1
Unique	AA		Asian		Hispanic		NHW	
AF	Count	%	Count	%	Count	%	Count	%
Singleton	33,348,856	76.5	29,033,386	70.1	44,486,085	67.5	61,301,483	76.0
<0.1%	10,107,022	23.2	9,289,546	22.4	20,809,053	31.6	19,206,135	23.8
0.1%–1%	151,690	0.3	2,916,818	7.0	644,326	1.0	133,697	0.2
1%–5%	-	-	157,348	0.4	1967	0.0	-	0.0
>5%	-	-	787	0.0	1	0.0	-	0.0

Abbreviations: AA, African American; ADSP, Alzheimer's Disease Sequencing Project; indel, insertion and deletion; NHW, non-Hispanic White; SNV, single nucleotide variant.

was generated using GATK4.1.1 on all SNVs and short indels. We identified 322,757,476 bi-allelic SNVs and 24,005,724 bi-allelic indels on autosomes, comprising 83% of the original GATK output, which will be the focus of this article. R4 data also contains 54,425,255 multi-allelic SNVs and indels on autosomes. A variant passes QC if it has a GATK "FILTER" = PASS or is in tranche $\geq 99.8\%$, read depth (DP) > 10 and genotype quality (GQ) > 20, genotype call Rate $\geq 80\%$, and is supported by < 500 reads (Method—ADSP Variant QC protocol in supporting information). Average call rate of the variants is high (97.0%). In GATK, a "tranche" represents a subset of variants categorized by their sensitivity to a truth set, allowing analysts to balance sensitivity and specificity when filtering variants. Tranches are defined by specific thresholds (e.g., 90, 99, or 99.9), in which higher tranche values correspond to higher specificity and a higher true positive rate, while lower tranche values capture more true positives but may also result in more false positives. This trade-off reflects a decrease in quality for the lower tranches due to the higher false positive rate. Additionally, the genotype call rate refers to the proportion of genotypes per marker that have non-missing data. Details for other quality of the variants can be found in Table S3 in supporting information.

The ADSP QC protocol flagged 92.94% of autosomal variants, 299,620,924 SNVs, and 22,674,845 indels, as high quality. Of the four major ethnicities, African American ($N = 5260$), Asian ($N = 2787$), Hispanic ($N = 10,972$), and non-Hispanic White ($N = 13,969$), there are 101,227,106 (94,371,761 SNVs and 6,855,642 indels) for African American, 69,338,361 (64,882,876 SNVs and 4,455,485 indels) for Asian, 132,424,746 (123,631,863 SNVs and 8,792,883 indels) for Hispanic, and 135,672,855 (126,113,083 SNVs and 9,559,772 indels) for non-Hispanic Whites (Table 1, top) respectively.

Regarding allele frequency (AF), 52.53% of variants are singletons, followed by 40.62% rare variants with AF < 0.1%, 3.72% with AF 0.1%

to 1%, 1.29% with AF 1% to 5%, and 1.84% with AF > 5%. The distribution of variants across AF ranges is consistent across ethnicities, with singletons comprising close to or > 50% of the variants. African American, Asian, and Hispanic groups (13.16%, 11.61%, and 7.53%, respectively) have a higher proportion of variants with AF > 1% compared to the overall dataset and the non-Hispanic White group (5.74% and 3.13%, respectively; Table 1, top).

In terms of ethnic specific variants, there are 43,607,568 (40,417,070 SNVs and 3,190,498 indels), 41,397,885 (38,544,025 SNVs and 2,853,860 indels), 65,941,432 (61,362,910 SNVs and 4,578,522 indels), 80,641,315 (7,447,578 SNVs and 6,165,529 indels) variants for African American, Asian, Hispanic, and non-Hispanic Whites, respectively. Most of the ancestral specific variants are rare variants (AF < 0.1%, Table S4 in supporting information; Table 1, bottom).

We compared the ADSP R4 bi-allelic variants to gnomAD³⁸ (Method—Comparison of gnomAD in supporting information). Of the ADSP R4 variants, 62.39% of SNVs and 57.33% of indels are reported in gnomAD. In terms of AF, 99.89% of variants with AF > 5%, 99.80% with 1% < AF $\leq$ 5%, 99.74% with 0.1% < AF $\leq$ 1%, 82.76% with AF $\leq$ 0.1%, and 41.08% of singletons are present in the gnomAD database.

3.5 | Annotation of genetic variants

The official ADSP annotation pipeline³⁹ was used to annotate all 347 million variants (SNVs and indels; Methods—Variant annotation protocol in supporting information; Figure 3A). Functional impact of variants was accessed using snpEff²⁹ (Methods—LOF analyses in supporting information). We identified 224,594 high-impact LoF variants: frameshift (39%), stop gained (27%), splice donor (16%), splice

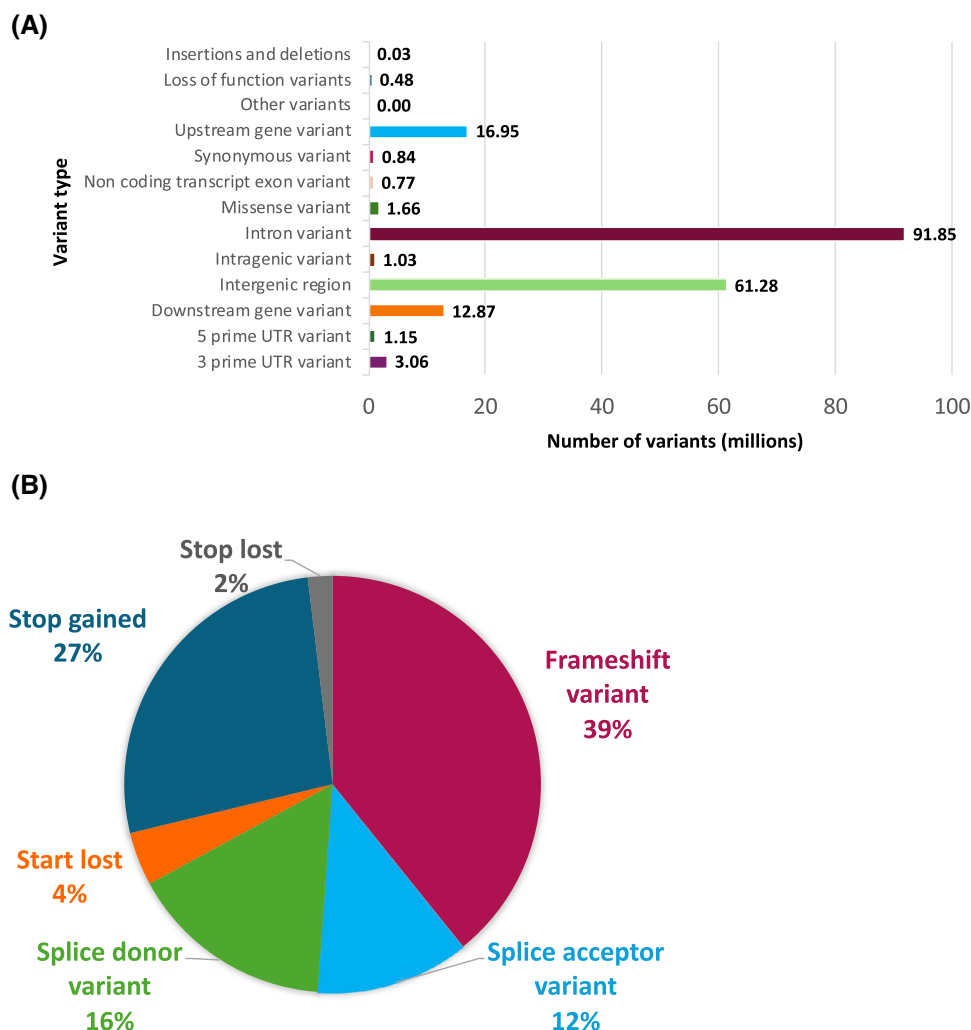


FIGURE 3 The distribution of variant types across the genome, with a specific focus on high-risk loss-of-function variants. A, Bar chart depicting the breakdown of the total number of variants across the genome, categorized by genomic annotation as follows: insertions and deletions, loss-of-function variants, upstream gene variants, synonymous variants, non-coding transcript exon variants, missense variants, intron variants, intragenic variants, intergenic variants, downstream variants, 5 prime UTR variants, and 3 prime UTR variants. B, The distribution of 224,594 loss-of-function variants is further broken down into the following categories: frameshift (39%), stop gained (27%), splice donor (16%), splice acceptor (12%), start lost (4%), and stop lost (2%). UTR, untranslated region.

acceptor (12%), start lost (4%), and stop lost (2%) across 22,710 genes (Figure 3B). Among these, 1295 genes were found to be intolerant to protein-truncating LoF variants, indicated by a Loss Intolerance Probability (pLI) score of 1, suggesting the critical importance of these genes. We also provide annotation for all bi-allelic variants using FAVOR (Methods—FAVOR annotation protocol in supporting information); 27.45 million variants of such are with CADD (phred score) of ≥ 20 , with > 63.64 million variants lying in super-enhancer regions.

3.6 | Structural variants in ADSP R4 samples

We applied the same protocol developed for the ADSP round 3 (R3) dataset⁴⁰ to the R4 dataset (Methods—Structural variant calling protocol in supporting information). Individual Manta and Smoove

callsets were initially merged for each sample, and then all samples were combined using SVIMMER (v0.1). GraphTyper (v2.7) was subsequently applied to the merged VCF for SV joint genotyping. Notably, only SVs > 10 Mbp were filtered. The final callset consists of a total of 6,796,267 SVs, including 4,101,354 deletions, 726,560 duplications, 558,860 insertions, and 1,409,493 inversions. Because an SV can be associated with multiple joint genotyping models, such as AGGREGATED, BREAKPOINT, BREAKPOINT1, BREAKPOINT2, and COVERAGE in GraphTyper2, some SVs appeared multiple times in the R4 SV VCF. After consolidating these models, the total number of unique SVs was reduced to 2,208,044, comprising 1,367,118 deletions, 184,367 duplications, 186,290 insertions, and 470,269 inversions.

On average, 15,640 high-quality SVs were identified per sample, including 7813 deletions, 1574 duplications, 6246 insertions, and 7

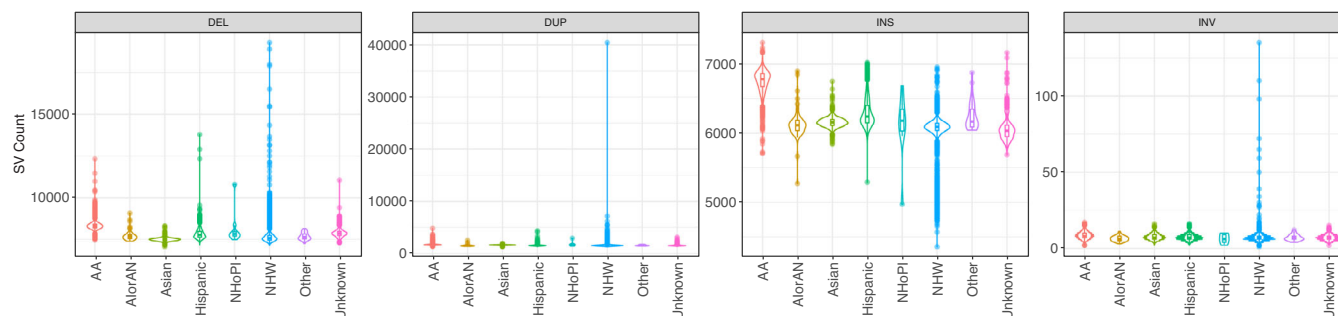


FIGURE 4 Comparison of the number of SVs called in the ADSP R4 dataset across different reported ethnicities. SVs can be categorized into four different types of SVs: deletion (DEL), duplication (DUP), inversion (INV), and insertion (INS). ADSP, Alzheimer's Disease Sequencing Project; SV, single variant.

inversions. Similar to the patterns observed with SNVs and indels, African American samples exhibited a higher number of SVs compared to other groups (Figure 4), consistent with previous studies indicating that the out-of-Africa bottleneck reduced genetic variation in non-African populations.⁴¹

3.7 | LD reference panel from ADSP R4 data

Starting with the ADSP Integrated Phenotypes list, which includes 32,236 samples (5096 African American, 2777 Asian, 10,438 Hispanic, 12,692 non-Hispanic White, and 1233 others), we constructed panels for African American, Asian, Hispanic, and non-Hispanic Whites separately. The panels were built using both SNVs and indels. The number of variants included in each panel was as follows: 46,462,895 variants (43,494,096 SNVs and 2,968,799 indels) for African Americans, 25,779,737 variants (24,235,526 SNVs and 1,544,211 indels) for Asians, 57,683,258 variants (54,000,742 SNVs and 3,682,516 indels) for Hispanics, and 44,629,226 variants (41,794,882 SNVs and 2,834,344 indels) for non-Hispanic Whites.

We performed emerald³⁴ with the following parameters, $-mac > 5$, $-threshold 0.2$, and $-window 5000000$. Each segment was analyzed by 5Mb window with 3Mb overlapping, then we concatenated all segments, removing duplicate records. As a result, we identified 3,153,513,864, 1,795,829,862, 4,990,587,680, and 3,205,008,552 pairs of variants for the African American, Asian, Hispanic, and non-Hispanic White samples, respectively. Among these, the proportion of pairs with $R^2 > 0.8$ was 6.3% for African American, 11.0% for Asian, 5.7% for Hispanic, and 9.2% for non-Hispanic White. For $R^2 > 0.2$, the proportions were 33.5%, 45.4%, 32.4%, and 40.9%, respectively.

3.8 | Harmonized phenotypic data for 28,000 ± participants with WGS

The ADSP-PHC was established to unify detailed endophenotype data from various cohort studies. The group collaborates with ADSP to ensure high-quality phenotype harmonization across multiple

domains, and document data availability and harmonization processes. Currently, available phenotypes from the ADSP-PHC include autopsy measures of neuropathology, fluid biomarkers of AD neuropathology, positron emission tomography measures of amyloid and tau pathology, structural brain imaging using magnetic resonance imaging, diffusion tensor imaging, longitudinal measures of cognition, and cardiovascular risk factor data. Harmonization methods are detailed in the Methods—ADSP Phenotype Harmonization Consortium (ADSP-PHC) section in supporting information.

To ensure the highest quality harmonization is conducted, the ADSP-PHC harmonizes all available phenotypic data, regardless of sequencing status, which the research community can access directly from each cohort. ADSP-PHC deliverables are then subset to participants with available sequencing data. To increase the value and usage of the ADSP dataset, the ADSP-PHC has selected > 9000 phenotypes across 10 domains, expanding on the three domains released previously (NIAGADS ng00067.v9). Figure 5 summarizes the harmonized data availability for more than 15,927 participants with WGS and harmonized phenotypic data in NIAGADS. A data availability and explorer tool are available online via <https://vmacdata.org/adsp-phc>.

3.9 | Summary of data files shared in this collection

All the described R4 data have been released in NIAGADS DSS (<https://dss.niagads.org/>). These include individual level CRAMs, gVCFs, and SV VCFs, as well as aggregated files including the joint-genotyped VCFs from 35,014 unique individuals. We offer alternative solutions for users who may not require VCFs with detailed information for their analyses. These options include genotype information, full quality metrics, or ADSP QC details, and are organized by bi-allelic and multi-allelic variants. Additionally, some of these files are available in the GDS,²¹ an alternative format to VCF designed specifically for R users. In addition, we provide sequencing methods, quality data metrics, variant metrics, phenotypes, and readmes along with these data files. Annotation and LD reference panel files are available in open access. We summarize these files by features and file size in Table 2.

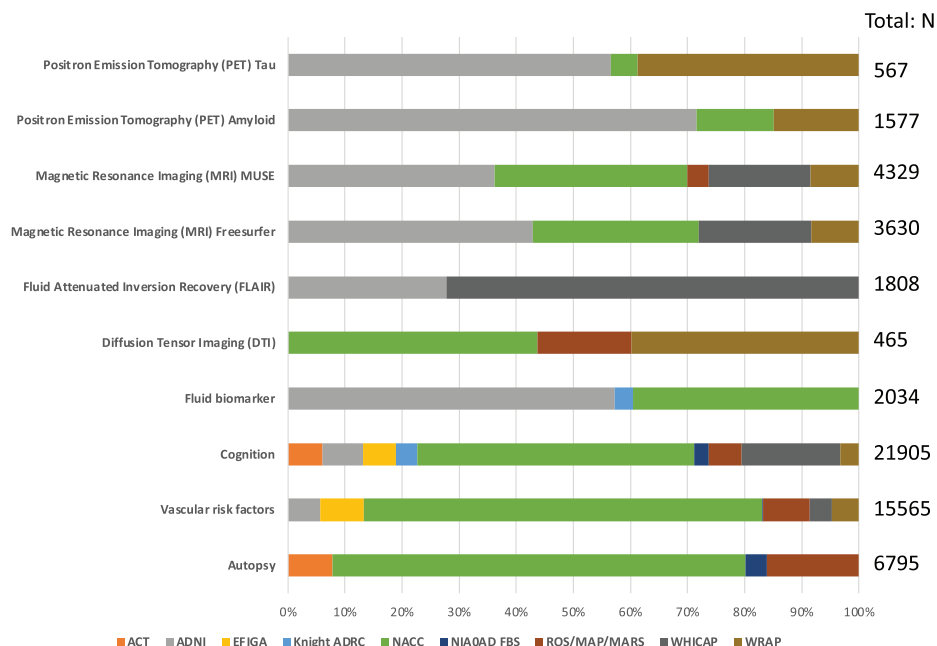


FIGURE 5 ADSP-PHC Release (ng00067.v11) Sample sizes ("N" on the right) reflect individuals with ADSP sequencing data in R4. ADSP, Alzheimer's Disease Sequencing Project; PHC, Phenotype Harmonization Consortium.

TABLE 2 ADSP R4 released file set. All files are available under <https://dss.niagads.org/datasets/ng00067/>. Both individual (CRAMs, gVCFs, SV VCFs, and phenotypes) and summary level files (VCFs, GDS, annotation files) are available. Annotation and LD reference panel files are also available in NIAGADS Open Access Data Portal <https://dss.niagads.org/open-access-data-portal/>.

Descriptions	Genotype quality information	ADSP QC info	With multiallelic
Individual level			
CRAMs	–	–	–
gVCFs	–	–	–
SV Manta VCFs	–	–	–
SV Smoove VCFs	–	–	–
Phenotypes	–	–	–
Summary level			
Preview VCFs	Full	No	Yes
Preview compact VCFs	Partial	No	Yes
Preview compact filtered VCFs	Partial	No	Yes
Fully QC-ed VCFs	Full	Yes	No
Fully QC-ed compact filtered VCFs	Partial	Yes	No
Fully QC-ed GDS	Partial	Yes	No
Annotation	–	–	–
LD reference panel	–	–	–

Abbreviations: Alzheimer's Disease Sequencing Project; GDS, Genomic Data Structure; indel, insertion and deletion; LD, linkage disequilibrium; NIAGADS, National Institute on Aging Genetics of Alzheimer's Disease Data Storage Site; QC, quality control; SV, structural variant; VCF, variant call format.

3.10 | Browser of variants and annotations of diversified samples

To allow users to explore the ADSP R4 genotypes without downloading the data, we provide users with two different visualization options. First, VariXam (<https://varixam.niagads.org/>) allows users to check the alleles and QC quality of any genetic variants in any callset

(this release R4 and earlier ADSP releases R1, R2, and R3; Figure 6A). Users can query by the SNV/gene (IDs or genomic coordinates) level or view all variants in a particular genomic region (Methods—VariXam in supporting information).

Second, the NIAGADS Alzheimer's GenomicsDB³⁵ (<https://www.niagads.org/genomics>) allows users to explore the variants and annotations in a broader genomic context (Methods—NIAGADS Alzheimer's

(A)

VariXam is an aggregated database and a variant browser that shows genomic variants detected on whole-genome/whole-exome sequence (WGS/WES) data of the [Alzheimer's Disease Sequencing Project \(ADSP\)](#). The variants were processed by [VCPA \(Variant Calling Pipeline and data management tool\)](#) and released by [NIAGADS](#). The database currently includes variants of R1 5K WGS, R2 20K WES, R3 17K WGS, and R4 36K WGS. The human reference genome used is GRCh38.

ADSP and NIAGADS are sponsored by the National Aging Institute (NIA) and aim to improve our understanding and provide insight of genetic factors of Alzheimer's Disease.

Enter the region, gene or rsID and click the search icon to see autosomal variants

APOE



Examples - Gene: [APOE](#) - Chromosome Position: [5:17100035](#) - Chromosome Region: [19:1593000-1594000](#) - rsID: [rs146621479](#)

Showing results for: **APOE**

Chromosome	Position	Ref Allele	Alt Allele	Project	Filter	Multiallelic	rsID
19	44907814	G	A	R2 20K WES	pass	No	
19	44909260	C	T	R4 36K WGS	pass	No	
19	44908288	G	C	R4 36K WGS	pass	No	
19	44906961	A	G	R4 36K WGS	pass	No	
19	44908102	T	C	R3 17K WGS	pass	No	
19	44909284	A	G	R2 20K WES	pass	No	
19	44905910	C	G	R2 20K WES	pass	No	rs440446
19	44907545	C	T	R4 36K WGS	pass	No	
19	44905806	C	T	R4 36K WGS	pass	No	

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(B)

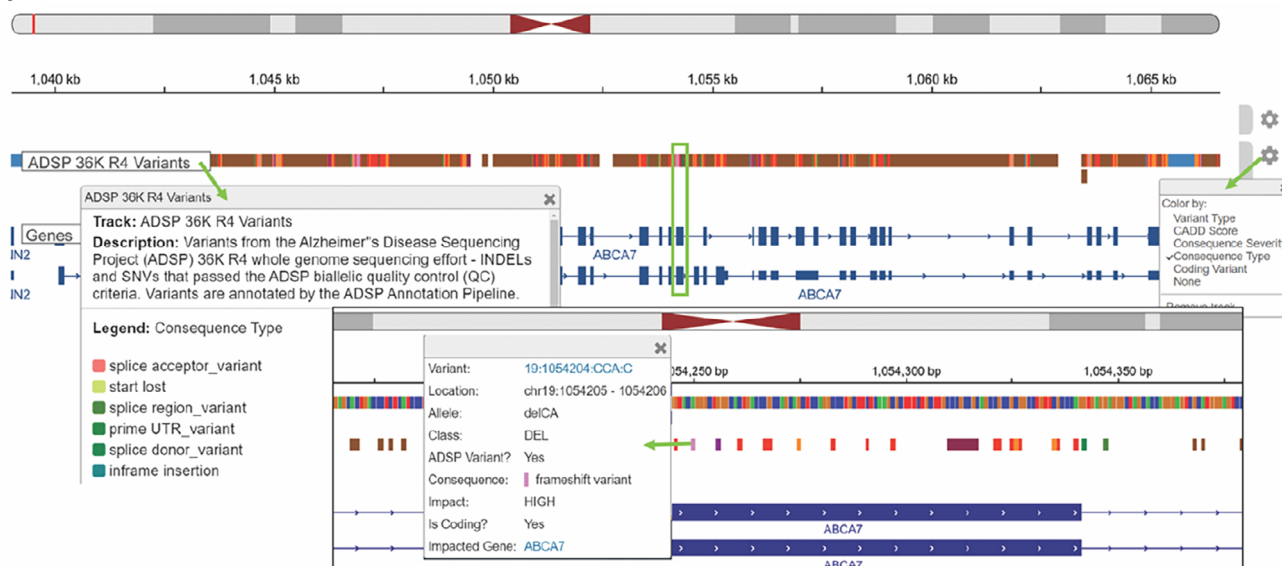


FIGURE 6 Browser of variants and annotations of diversified samples. A, VariXam interface. A variant browser displaying all genomic variants identified in the ADSP whole genome and exome data across releases. The figure below shows the search results of *APOE*. Accessible at: <https://varixam.niagads.org/>. B, The R4 variants can be visually inspected as a track on the NIAGADS Genome Browser. The track displays annotated short INDELS and SNVs that passed the biallelic QC criteria. Track annotations include the most severe variant consequences and

GenomicsDB in supporting information). It provides detailed reports of genetic associations from NIAGADS-hosted GWAS summary statistic datasets in the context of genes and annotated variant records. ADSP R4 variants are flagged, and can be filtered by the most severe consequence predicted by the ADSP annotation pipeline.³⁹ Variant reports also include the full ADSP annotation results (all predicted consequences, CADD³⁰ deleteriousness, LoF,³¹ and FAVOR annotations³¹), allele frequencies (including gnomAD³⁷ and 1000Genomes⁴²), and links to related web resources. The GenomicsDB genome browser provides an ADSP R4 variant track that can be recolored based on annotations (e.g., coding vs. non-coding variants, consequence type). This track can be compared to the summary statistic data or other tracks in GenomicsDB (Figure 6B).

4 | DISCUSSION

The R4 WGS dataset is the largest and most diverse whole genome data collection for AD to date. Numerous studies using the ADSP WGS data in a smaller scale (previous R3 release, 46% of the current sample size, less diversified) have led to multiple findings in AD genetics,^{43–60} including but not limited to the discovery of (1) new AD genes *PLEC*, *UTRN*, *TP53*, and *POLD1*⁴⁵ using a novel approach, GeneEMBED, designed for studying gene interactions; *DLG2* and *DTNB* via rvGWAS on family datasets⁴³; (2) rare genetic variants contributing to AD risk⁴⁴; (3) sex-specific loci identified in family-based designs⁴⁸; (4) novel AD risk loci on 13q33.3 via admixture mapping analyses in Caribbean Hispanic populations⁵⁶; and (5) novel AD associations in Ashkenazi Jews with variants that are exceedingly rare or absent in other European ancestry populations.⁶¹ The ADSP WGS data have also enabled the further study of *APOE* in different ancestries^{49,50,53,58,59} and identification of associations of AD risk in multiple population groups with human viruses detected among unmapped reads in the WGS data.⁶² Finally, these data have led to new insights in early-onset Alzheimer's (EOAD),⁵¹ enabled the detection of nuclear and mitochondrial copy number variations and structural variants and their association with LOAD,^{46,52,63} as well as the generation of ancestry-specific polygenic risk score in the Amish population⁴⁷ and the development of a pipeline for calling mitochondrial sequence variants and haplogroups.⁶⁴

We faced several challenges in this project. First, samples are recruited from independent cohorts/projects of various study designs and sequencing experiments funded during the past 10 years, and challenges rise for GCAD to process and cumulatively integrate all genomic data. Unlike AllofUS⁶⁵ and UKBiobank,⁶⁶ the ADSP program has to

address the issue of heterogeneity in recruitment criteria, phenotype data collection protocols, and sequencing platform and configuration changes as sequencing technology and analysis best practices continue to evolve. Subtle batch effects may persist in our datasets even after we process all sequence data and perform thorough QCs during the data harmonization process at sample, variant, and phenotype levels.

Residual batch effects can introduce biases, confounding the relationship between genetic variants and phenotypes if not properly addressed. This is especially relevant for projects like ADSP, in which factors such as case/control counts and sequencing protocols vary across cohorts, leading to potential spurious associations driven by batch effects. Therefore, it is suggested that analysts generate Q-Q plots after performing genetic associations to check for genetic inflation. If no inflation is observed, batch effects are likely minimal and absorbed by the PCA plot, ensuring they won't interfere with subsequent analyses.

Batch effects can also distort population structure and relatedness analyses, leading to incorrect conclusions about genetic ancestry or stratification. This, in turn, can affect downstream analyses like imputation or polygenic risk score calculations. One way to evaluate the quality of PCA plots is by comparing them across cohorts with similar genetic makeup or using only technical/biological replicates, which should ideally cluster together.

Moreover, residual batch effects can impact cross-cohort comparisons, meta-analyses, or replication studies, particularly in machine learning models. Unlike genetic association regression models, machine learning models can't easily account for covariates, making them prone to learning batch-specific artifacts rather than true biological signals, potentially leading to overfitting or poor generalizability.

Second, compared to previous releases (R1 and R3 for genomes), this ADSP R4 dataset more than doubled the sample size (35,014 vs. 16,285) with the biggest growth in the Hispanic and Latino group (Figure S5 in supporting information), yet the Asian population is still significantly underrepresented. These gaps must be addressed if we are to fully understand the distribution and effect of human genetic variation in AD. Power analyses show we still need 18,500 cases and 18,500 controls per ancestry group to gain enough power for detecting variants with MAF of 0.005. This power calculation refers to the expected sample size we would need to detect (associate) a variant at genome-wide significance ($p \leq 5 \times 10^{-8}$) with a disease AF of 0.005 assuming the disease has a prevalence of 10% and a genotype relative risk ≥ 1.68 .

In conclusion, we have shown that the ADSP R4 resource offers researchers in the neurodegenerative field multiple tools to explore

consequence impacts predicted by the ADSP annotation pipeline and mappings to dbSNP refSNP identifiers. The track settings menu can be used to recolor the variants based on various annotations; the legend (made available by clicking on the track name) will update accordingly. Users can zoom into regions of interest (here, the green rectangle highlights the region displayed in the close-up inset) to view sequence information and click on individual variants for a brief summary of the annotations. Full annotation results can be browsed by following the link to the GenomicsDB record for the variant. ADSP, Alzheimer's Disease Sequencing Project; *APOE*, apolipoprotein E; INDEL, insertion and deletion; NIAGADS, National Institute on Aging Genetics of Alzheimer's Disease Data Storage Site; QC, quality control; SNV, single nucleotide variant.

and analyze the genetic variations associated with these diseases. We anticipate that these data will significantly influence nearly all ongoing studies on common and rare variants in AD, with an even greater impact as additional samples from diverse populations become available. Moving forward, we plan to provide annotations for the identified SVs further empowering researchers who depend on such resources to conduct fine-mapping and other post-genome-wide genetic analyses. We will also provide machine learning based annotations such as those generated by AlphaMissense or SpliceAI to enhance the breadth of future analyses on these data.

The ADSP program needs to share data in a manner that supports the privacy and consent preferences of participants. Members of the scientific community can access most ADSP resources (individual-level sequence files, phenotype data, VCFs) through the NIAGADS DSS platform (dbGAP-like). Some companion data (e.g., annotation LD reference panel) are open access, while VariXam variant server and NIAGADS genomicsDB provide a preview of the data without any restrictions.

The ADSP R4 dataset of > 347 million variants is available to qualified investigators at <https://dss.niagads.org/datasets/ng00067/>.

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CONFLICT OF INTEREST STATEMENT

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CODE AVAILABILITY

VCPA code: <https://bitbucket.org/NIAGADS/vcpa-pipeline/src/master/>

SV related code: <https://github.com/DecodeGenetics/svimmer>, <https://github.com/DecodeGenetics/graph typer>

QC code: https://bitbucket.org/Taha_Iqbal_UPenn/gcad_vcf-qc_public/

DATA AVAILABILITY STATEMENT

NIAGADS DSS is accessible at: <https://dss.niagads.org/>

VariXam is accessible at: <https://varixam.niagads.org/>

GenomicsDB is accessible at: <https://www.niagads.org/genomics/app>

CONSENT STATEMENT

All human subjects provided informed consent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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